SSC MTS 2024 question paper with answers

14 October 2024 · Shift 3

81 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 9 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 16 questions

Q.1

The outer circumference of the top of a large hemispherical iron bowl is 565.2 cm and the bowl is 6 cm thick. What will be the volume (in cm^3) of the iron required to make this bowl? (Use $\pi = 3.14$.)

(A) 2,85,312.96
(B) 2,85,212.96
(C) 2,85,232.96
(D) 2,85,132.96

Q.2

A shop keeper allows a 25% discount and gains 20%. How much of the percentage is above the Marked Price on the Cost Price?

(A) 60%
(B) 50%
(C) 80%
(D) 20%

Q.3

Two trains are running in opposite directions at the same speed. The length of each train is 180 m. If they cross each other in 12 seconds, what is the speed of each train (in km/h)?

(A) 36
(B) 54
(C) 63
(D) 45

Q.4

If the volume of a sphere is divided by its surface area, the result is 54 cm. The radius of the sphere is:

(A) 111 cm
(B) 166 cm
(C) 182 cm
(D) 162 cm

Q.6

Three candidates contested an election and obtained 1136, 7636 and 11628 votes, respectively. What percentage of the total votes did the losing candidate (last one) get? (Correct to two decimal places)

(A) 6.34%
(B) 3.89%
(C) 7.89%
(D) 5.57%

Q.7

The average age of a class of 65 boys is 14 years. The average age of 20 of them is 14 years, and the average age of another 15 boys is 12 years. What will be the average age (in years) of the remaining boys?

(A) 16
(B) 17
(C) 14
(D) 15

Q.8

A thief steals a bike at 12.40 p.m. and drives it at 45 km/h. The theft is discovered at 1 p.m. and the owner sets off in a car at 65 km/h in the direction the thief was heading. The time at which the owner can catch the thief is:

(A) 1:45 p.m.
(B) 2:00 p.m.
(C) 2:10 p.m.
(D) 1:30 p.m.

Q.9

Find the simple interest on ₹550 for 8 months at 3 paise per rupee per month?

(A) ₹134
(B) ₹132
(C) ₹130
(D) ₹131

Q.10

A man purchased 3 bags of cement at the rate of ₹2,200 each, 4 bags of cement at ₹2,550 each and 5 bags of cement at the rate of ₹3,120 each. What is the average cost of one bag of cement?

(A) ₹2,700
(B) ₹2,800
(C) ₹2,623
(D) ₹2,805

Q.11

The average height of a group of 50 students is 175 cm. If another group of 40 students with average height of 180 cm is included, what is the average (in cm, to the nearest cm) of the combined group?

(A) 178
(B) 177
(C) 174
(D) 176

Q.12

Find the simple interest (in ₹) on ₹2000 as a sum borrowed at 6% per year rate of interest for 3 years.

(A) 320
(B) 400
(C) 360
(D) 300

Q.14

A shopkeeper sells goods at the same price rate for which he has bought them, but gives 20% less quantity to the customer. Find the profit percentage.

(A) 25%
(B) 20%
(C) 15%
(D) 30%

Q.15

Find the simple interest (in ₹) on ₹3500 as a sum borrowed at 7% per year rate of interest for 3 years.

(A) 695
(B) 775
(C) 735
(D) 675

Q.18

In what ratio wheat at ₹38 per kg should be mixed with wheat at ₹26 per kg so that on selling the mixture at ₹39 per kg, there would be a profit of 30%?

(A) 1 : 3
(B) 1 : 2
(C) 2 : 3
(D) 3 : 4

Q.19

What is the HCF of 512 and 296?

(A) 4
(B) 8
(C) 37
(D) 32

Q.20

A group of 9 people can survive on ₹8,100 for 30 days. How long can a group of 7 people survive on ₹8,400?

(A) 40 days
(B) 43 days
(C) 25 days
(D) 31 days

Reasoning Ability and Problem Solving 16 questions

Q.1

Seven friends, Monu, Abhay, Vicky, Chitti, Deva, Hamza and Sanjay, are attending Hamza's birthday party and sitting in a circular arrangement facing the centre(but not necessarily in the same order). Monu is an immediate neighbour of Abhay and Deva. Abhay is second to the left of Hamza. Sanjay is one of the immediate neighbours of Hamza. Chitti is an immediate neighbour of both Deva and Vicky. Who is sitting third to the left of Monu?

(A) Hamza
(B) Deva
(C) Vicky
(D) Sanjay

Q.2

MAZE is related to QEDI in a certain way based on the English alphabetical order. In the same way, YMLQ is related to CQPU. To which of the following is OCBG related, following the same logic?

(A) SFGK
(B) SGFK
(C) GKSF
(D) SGKF

Q.3

181 is related to 417 following a certain logic. Following the same logic, 149 is related to 385. To which ofthe following is 262 related, following the same logic? (NOTE: Operations should be performed on the whole numbers, without breaking down the numbersinto its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1and 3 is not allowed.)

(A) 465
(B) 437
(C) 498
(D) 474

Q.5

In a code language, 'RAMESH' is written as 'UAPEVK' and 'TOM' is written as 'WOP'. How will 'MEGHANA' be written in the same language?

(A) QDJKAQA
(B) PEJKAQA
(C) QEJKAQA
(D) PDJKAQA

Q.8

EKQW is related to JPVA in a certain way based on the English alphabetical order. In the same way, BHNT is related to GMSX. To which of the following is PVBH related, following the same logic?

(A) UBGL
(B) UABL
(C) UAGL
(D) UCGL

Q.9

What should come in place of the question mark (?) in the given series? 89 92 98 107 119 ?

(A) 114
(B) 124
(C) 144
(D) 134

Q.10

What will come in place of the question mark '?' in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 13 + 4 × 2 ÷ 5 – 2 = ?

(A) 7
(B) 5
(C) 3
(D) 2

Q.11

Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g., 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) (25, 8, 400) (12, 14, 336)

(A) (19, 8, 152)
(B) (16, 21, 672)
(C) (22, 9, 198)
(D) (17, 11, 189)

Q.12

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: All hens are chickens. All chickens are eggs. Conclusions: (I) All hens are eggs. (II) All eggs are chickens.

(A) Only conclusion I follows
(B) Neither conclusion I nor II follows
(C) Both conclusions I and II follow
(D) Only conclusion II follows

Q.13

What will come in place of the question mark (?) of the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 22 × 11 + 13 – 12 ÷ 2 = ?

(A) 13
(B) 35
(C) 22
(D) 17

Q.14

In a code language, 'ARVL' is coded as 'FOAI' and 'DOXY' is coded as 'ILCV'. How will 'MPSB' be coded in that language?

(A) QNWZ
(B) QMWY
(C) RNXZ
(D) RMXY

Q.15

Six players Jadeja, Ashwin, Karthik, Murli, Sarfaraz and Virat are sitting around a circular table, facing the centre (but not necessarily in the same order) . Virat is sitting to the immediate right of Jadeja. Only Sarfaraz is sitting between Virat and Murli. Jadeja is an immediate neighbour of Ashwin. Who is immediate neighbour of both Sarfaraz and Karthik?

(A) Ashwin
(B) Murli
(C) Jadeja
(D) Virat

Q.16

In a certain code language, 'barks a lot' is written as 'qw er ui' and 'the dog barks' is writtenas 'op ji er'. How is 'barks' writtenin that language?

(A) qw
(B) ji
(C) ui
(D) er

Q.18
What should come in place of the question mark (?) in the given series based on the English alphabetical order? SDW, OHS, KLO, GPK ,?

(A) CSG
(B) BSG
(C) CTG
(D) BTG

Q.19
In a certain code language, 'BEAM' is coded as '5791' and 'MINT' is coded as '2467'. What is the code for 'M' in the given code language?

(A) 6
(B) 2
(C) 7
(D) 1

Q.20
Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: Some possums are bisons. No possum is a wolf. Conclusions: [1] All wolves can be bisons. [2] All bisons are wolves.

(A) Only conclusion [1] follows
(B) Both conclusions [1] and [2] follow
(C) Neither conclusion [1] nor [2] follows
(D) Only conclusion [2] follows

General Awareness 25 questions

Q.1
Which of the following is an autobiography of LK Advani?

(A) Dare to Dream
(B) My Country My Life
(C) Exam Warriors
(D) Wings of Fire

Q.2
According to Global Multidimensional Poverty Index 2022, what was the population of the poor in India in 2020?

(A) 428.9 million
(B) 128.9 million
(C) 328.9 million
(D) 228.9 million

Q.3
The strength of Supreme Court has increased to how many judges as of 13 Feb 2023?

(A) 30
(B) 34
(C) 26
(D) 28

Q.4
Who among the following social reformers wrote Brahmo Covenant which was a statement of the creed of the Brahmo Samaj?

(A) Raja Ram Mohan Roy
(B) Keshab Chandra Sen
(C) Debendranath Tagore
(D) Ishwar Chandra Vidyasagar

Q.5
Which sector uses the groundwater maximally?

(A) Educational
(B) Industrial
(C) Domestic
(D) Agriculture

Q.6

Who among the following Governor Generals of India wrote about the predicament of Indian weavers, “The bones of the cotton weavers are scattered on the plains of India”?

(A) Lord Amherst
(B) Sir John Shore
(C) Lord William Bentinck
(D) Lord Minto

Q.7

Who became the first woman to win two gold medals in a single Para Asian Games event in 2023?

(A) Sheetal Devi
(B) Disha Dhayal
(C) Neema Rani
(D) Sarita Devi

Q.8

Which of the following crops takes almost a year to grow?

(A) Ragi
(B) Groundnut
(C) Soyabean
(D) Sugarcane

Q.9

Wasifuddin Dagar was associated with which of the following classical music genres?

(A) Ghazal
(B) Dhrupad
(C) Tarana
(D) Khyal

Q.10

In which of the following states is the famous International Sand Art Festival organised, where artists get enthralled by the diverse thoughts and their elucidation on sand?

(A) Assam
(B) Bihar
(C) West Bengal
(D) Odisha

Q.11

Which of the following options is an example of a decomposer in an ecosystem?

(A) Herbivore
(B) Carnivore
(C) Fungus
(D) Producer

Q.12

Sama Chakeva is a folk dance based on the mythological story of the siblings Sama and Chakeva, in which the brother goes to an extreme extent to save his sister. With which State of India is this folk dance associated?

(A) Bihar
(B) Assam
(C) Punjab
(D) Haryana

Q.13

In the periodic table, what is the trend for atomic radius as you move from left to right across a period (horizontal row)?

(A) Atomic radius increases
(B) Atomic radius remains constant
(C) Atomic radius decreases
(D) Atomic radius exhibits irregular behaviour

Q.14

Which state received the Sportstar ACES Best State for the Promotion of Sport award in 2024?

(A) Odisha
(B) Punjab
(C) Haryana
(D) Tamil Nadu

Q.15
Who amongst the following said that the government fiscal policy should be used to stabilise the level of output and employment?

(A) JM Keynes
(B) Alfred Marshall
(C) Adam Smith
(D) Lionel Charles Robbins

Q.16
The Megalithic site _____ is a site on the river Ghod, a tributary of the Bhima.

(A) Inamgaon
(B) Hallur
(C) Pottru
(D) Hire benakal

Q.17
The part of a temple wherein the main deity of the temple is housed is known as the _____.

(A) Mandap
(B) Stupa
(C) Garbha Griha
(D) Shikara

Q.18
The black coating formed on silver jewellery is:

(A) AgCl
(B) Ag_2CO_3
(C) Ag_2S
(D) $AgNO_3$

Q.19
Which of the following white papers, released by NITI Aayog, is related to Public Health Surveillance in India?

(A) Vision 2035
(B) Arogya Mission
(C) Mission Karmayogi
(D) Dhruv Mission

Q.20
The 'Rule of Law' is a key feature of the Indian Constitution which is a doctrine ascribed to _____.

(A) Karl Marx
(B) Adam Smith
(C) Albert Venn Dicey
(D) Plato

Q.21
The Dudhsagar waterfall is a major tourist place of _____ State.

(A) Nagaland
(B) Chhattisgarh
(C) Goa
(D) Meghalaya

Q.22
When was the Lahore declaration signed between India and Pakistan?

(A) December 1991
(B) July 1948
(C) August 1998
(D) February 1999

Q.23
Which of following silver coins was introduced by Sultan Shamsuddin Iltutmish?

(A) Jittal
(B) Dinar
(C) Tanka
(D) Adali

Q.24
Gulabo Sapera, a legendry folk dancer from Kalbeliya tribe, Rajasthan, received Padma Shri in the year _____.

(A) 2016
(B) 2021
(C) 2017
(D) 2023

Q.25
The legislative powers in India are divided into three lists, which are the Union List, the State List and the _____.

(A) Exclusive List
(B) Concurrent List
(C) Federal List
(D) Regional List

English Language and Comprehension 24 questions

Q.1
Select the most appropriate option that can substitute the underlined segment in the given sentence. If there is no need to substitute it, select 'No substitution'. Commodity prices are the "dearer" in the present economic scenario.

(A) quite dear
(B) dearest
(C) No substitution
(D) dear

Q.2
Select the option that can be used as a one-word substitute for the given group of words. A person who abstains from any sort of alcoholic drinks

(A) Samaritan
(B) Termagant
(C) Veteran
(D) Teetotaller

Q.3
The following sentence has been divided into parts. One of them may contain an error. Select the part that contains the error from the given options. If you don't find any error, mark 'No error' as your answer. The students were diligently / studying for their exams / for weeks before you arrived.

(A) studying for their exams
(B) No error
(C) for weeks before you arrived
(D) The students were diligently

Q.4
Select the option with the correct spelling to replace the underlined word in the given sentence. Polluted air can be "dangeros", even if the pollutants are invisible.

(A) dangeres
(B) dangeris
(C) dangerous
(D) dangerus

Q.5
The following sentence has been divided into parts. One of them may contain an error. Select the part that contains the error from the given options. If you don't find any error, mark 'No error' as your answer. Alaska in the USA / is the colder / of all the states.

(A) of all the states.
(B) is the colder
(C) Alaska in the USA
(D) No error

Q.6
The following sentence contains an error. Select the option that correctly rectifies the error. His aunt who lives at America brought a lot of gifts for him and his sister.

(A) lot of gifts by him
(B) His aunt which lives
(C) and her sister
(D) in America brought a

Q.7

In the following sentence the underlined part contains an error. Alternatives to the underlined part are given as options. Select the correct alternative. Which flower do you like “best”?

(A) the best
(B) better
(C) good
(D) most good

Q.8

Select the most appropriate ANTONYM of the word given in brackets to fill in the blank. After the floods, Uttarakhand was in a _____(ravaged) condition.

(A) restored
(B) influenced
(C) ratified
(D) languid

Q.9

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. The artist believes that this new collection of paintings / has being / a masterpiece that / will redefine contemporary art.

(A) a masterpiece that
(B) will redefine contemporary art
(C) has being
(D) The artist believes that this new collection of paintings

Q.10

Select the option that can be used as a one-word substitute for the given group of words. Something that is so extraordinary or unbelievable that it's hard to believe it

(A) Exquisite
(B) Ordinary
(C) Horrible
(D) Incredible

Q.11

Select the most appropriate meaning of the underlined phrase. I should “look into my past” and decide what the next course of action should be.

(A) Disrespect
(B) Introspect
(C) Prospect
(D) Retrospect

Q.12

Select the most appropriate meaning of the underlined idiom. Sushant looked “blue in the face” after the marathon.

(A) happy
(B) exhausted
(C) sad
(D) good

Q.13

Select the INCORRECTLY spelt word.

(A) Express
(B) Excess
(C) Aresst
(D) Harass

Q.14

Select the most appropriate option that can substitute the underlined segment in the given sentence. If there is no need to substitute it, select 'No substitution required'. If she “had gone to the party”, she could have met Amit.

(A) No substitution required
(B) has gone to the party
(C) have went to the party
(D) had go to the party

Q.15

Select the most appropriate synonym of the underlined word. The team worked together to “execute” the challenging task.

(A) analyse
(B) accomplish
(C) abandon
(D) announce

Q.16

Identify the ANTONYM pair.

(A) Influence and Motivate
(B) Convince and Persuade
(C) Dissuade and Encourage
(D) Encourage and Support

Q.17

Select the most appropriate ANTONYM of the given word. Timid

(A) Humble
(B) Oppress
(C) Modest
(D) Bold

Q.18

Select the most appropriate meaning of the underlined idiom. She finally “spilled the beans” about the surprise party.

(A) Revealed a secret or disclosed information
(B) Prepared a bean dish
(C) Caused a mess
(D) Dropped a container of beans

Q.19

Select the most appropriate synonym of the underlined word. The students demonstrated their “excitement” by volunteering for community service.

(A) anxiety
(B) ignorance
(C) enthusiasm
(D) freedom

Q.20

Select the correct spelling of the underlined word in the given sentence. We do not “expet” Reema to follow all the rules from the first day of work.

(A) execpt
(B) expect
(C) expact
(D) expict

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. A brigantine is a two-masted sailing vessel (1)_____ fore-and-aft rigging on the mainmast and square rigging on the foremast. The pirates that (2)_____ the Mediterranean in the sixteenth century loved it for its speed and ease of handling. The brigantine (3)_____ into a true sailing vessel in the waters of northern Europe. Although the words began to be used interchangeably, it was (4)_____ from the fully square-rigged brig by its gaff-rigged mainsail. True brigantines, for (5)_____, were defined as having square topsails above a gaffed mainsail.

Select the most appropriate option to fill in blank number 1.

(A) predicting
(B) focusing
(C) featuring
(D) attending

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. A brigantine is a two-masted sailing vessel (1)_____ fore-and-aft rigging on the mainmast and square rigging on the foremast. The pirates that (2)_____ the Mediterranean in the sixteenth century loved it for its speed and ease of handling. The brigantine (3)_____ into a true sailing vessel in the waters of northern Europe. Although the words began to be used interchangeably, it was (4)_____ from the fully square-rigged brig by its gaff-rigged mainsail. True brigantines, for (5)_____, were defined as having square topsails above a gaffed mainsail.

Select the most appropriate option to fill in blank number 2.

(A) ravaged
(B) fixed
(C) integrated
(D) transpired

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. A brigantine is a two-masted sailing vessel (1)_____ fore-and-aft rigging on the mainmast and square rigging on the foremast. The pirates that (2)_____ the Mediterranean in the sixteenth century loved it for its speed and ease of handling. The brigantine (3)_____ into a true sailing vessel in the waters of northern Europe. Although the words began to be used interchangeably, it was (4)_____ from the fully square-rigged brig by its gaff-rigged mainsail. True brigantines, for (5)_____, were defined as having square topsails above a gaffed mainsail.

Select the most appropriate option to fill in blank number 3.

(A) discharged
(B) garnished
(C) submitted
(D) evolved

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. A brigantine is a two-masted sailing vessel (1)_____ fore-and-aft rigging on the mainmast and square rigging on the foremast. The pirates that (2)_____ the Mediterranean in the sixteenth century loved it for its speed and ease of handling. The brigantine (3)_____ into a true sailing vessel in the waters of northern Europe. Although the words began to be used interchangeably, it was (4)_____ from the fully square-rigged brig by its gaff-rigged mainsail. True brigantines, for (5)_____, were defined as having square topsails above a gaffed mainsail.

Select the most appropriate option to fill in blank number 5.

(A) instance
(B) location
(C) case
(D) mention

Answer key

Numerical and Mathematical Ability

Q.1 **(A)**	Q.2 **(A)**	Q.3 **(B)**	Q.4 **(D)**	Q.6 **(D)**	Q.7 **(D)**
Q.8 **(A)**	Q.9 **(B)**	Q.10 **(A)**	Q.11 **(B)**	Q.12 **(C)**	Q.14 **(A)**
Q.15 **(C)**	Q.18 **(B)**	Q.19 **(B)**	Q.20 **(A)**		

Reasoning Ability and Problem Solving

Q.1 **(C)**	Q.2 **(B)**	Q.3 **(C)**	Q.5 **(B)**	Q.8 **(C)**	Q.9 **(D)**
Q.10 **(B)**	Q.11 **(B)**	Q.12 **(A)**	Q.13 **(A)**	Q.14 **(D)**	Q.15 **(B)**
Q.16 **(D)**	Q.18 **(C)**	Q.19 **(C)**	Q.20 **(A)**		

General Awareness

Q.1 **(B)**	Q.2 **(D)**	Q.3 **(B)**	Q.4 **(C)**	Q.5 **(D)**	Q.6 **(C)**
Q.7 **(A)**	Q.8 **(D)**	Q.9 **(B)**	Q.10 **(D)**	Q.11 **(C)**	Q.12 **(A)**
Q.13 **(C)**	Q.14 **(D)**	Q.15 **(A)**	Q.16 **(A)**	Q.17 **(C)**	Q.18 **(C)**
Q.19 **(A)**	Q.20 **(C)**	Q.21 **(C)**	Q.22 **(D)**	Q.23 **(C)**	Q.24 **(A)**
Q.25 **(B)**					

English Language and Comprehension

Q.1 **(B)**	Q.2 **(D)**	Q.3 **(D)**	Q.4 **(C)**	Q.5 **(B)**	Q.6 **(D)**
Q.7 **(A)**	Q.8 **(A)**	Q.9 **(C)**	Q.10 **(D)**	Q.11 **(D)**	Q.12 **(B)**
Q.13 **(C)**	Q.14 **(A)**	Q.15 **(B)**	Q.16 **(C)**	Q.17 **(D)**	Q.18 **(A)**
Q.19 **(C)**	Q.20 **(B)**	Q.21 **(C)**	Q.22 **(A)**	Q.23 **(D)**	Q.25 **(A)**

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (A) 2,85,312.96

Outer radius = 565.2 ÷ (2 × 3.14) = 90 cm, so the inner radius is 84 cm. Volume of iron = (2/3)π($90^3 - 84^3$) = (2/3) × 3.14 × 1,36,296 = 2,85,312.96 cm^3.

Q.2 · Answer (A) 60%

Selling price = 1.2 × CP = 0.75 × MP, so MP = (1.2/0.75) × CP = 1.6 × CP: the marked price is 60% above the cost price.

Q.3 · Answer (B) 54

Crossing each other, the trains cover 180 + 180 = 360 m in 12 seconds at their combined speed of 30 m/s. Each runs at 15 m/s = 15 × 18/5 = 54 km/h.

Q.4 · Answer (D) 162 cm

$(4/3)\pi r^3 \div 4\pi r^2 = r/3 = 54$, so r = 162 cm.

Q.6 · Answer (D) 5.57%

The losing candidate, the one who came last, got 1,136 of the 1,136 + 7,636 + 11,628 = 20,400 votes: 1,136/20,400 × 100 = 5.57%.

Q.7 · Answer (D) 15

The class's total age is 65 × 14 = 910 years. The two groups account for 20 × 14 + 15 × 12 = 460, leaving 450 years for the other 30 boys: 450/30 = 15.

Q.8 · Answer (A) 1:45 p.m.

By 1 p.m. the thief is 20 minutes ahead: 45 × 1/3 = 15 km. The owner gains 65 – 45 = 20 km/h, so he needs 15/20 hour = 45 minutes: 1:45 p.m.

Q.9 · Answer (B) ₹132

3 paise per rupee per month is 3% a month: 550 × 3/100 × 8 = ₹132.

Q.10 · Answer (A) ₹2,700

3 × 2,200 + 4 × 2,550 + 5 × 3,120 = 6,600 + 10,200 + 15,600 = ₹32,400 for 12 bags: ₹2,700 a bag.

Q.11 · Answer (B) 177

(50 × 175 + 40 × 180)/90 = (8,750 + 7,200)/90 = 15,950/90 = 177.2, or 177 cm to the nearest cm.

Q.12 · Answer (C) 360

SI = 2,000 × 6 × 3/100 = ₹360.

Q.14 · Answer (A) 25%

He charges for 100 units but hands over 80, which cost him the price of 80: profit = 20/80 × 100 = 25%.

Q.15 · Answer (C) 735

SI = 3,500 × 7 × 3/100 = ₹735.

Q.18 · Answer (B) 1 : 2

For a 30% profit at ₹39, the mixture must cost 39/1.3 = ₹30 a kg. By alligation, ₹38 wheat : ₹26 wheat = (30 – 26) : (38 – 30) = 4 : 8 = 1 : 2.

Q.19 · Answer (B) 8

512 = 2^9 and 296 = $2^3 \times 37$, so the HCF is 2^3 = 8.

Q.20 · Answer (A) 40 days

₹8,100 lasts 9 people 30 days, so one person needs 8,100/(9 × 30) = ₹30 a day. ₹8,400 then lasts 7 people 8,400/(7 × 30) = 40 days.

Reasoning Ability and Problem Solving

Q.1 · Answer (C) Vicky
Facing the centre, left is clockwise. Abhay is second to Hamza's left, Monu sits between Abhay and Deva, Chitti between Deva and Vicky, and Sanjay beside Hamza, so clockwise they run Hamza, Sanjay, Abhay, Monu, Deva, Chitti, Vicky. Third to Monu's left is Vicky.

Q.2 · Answer (B) SGFK
Each letter moves 4 places forward (Z wraps round to D): MAZE → QEDI and YMLQ → CQPU. So OCBG → SGFK.

Q.3 · Answer (C) 498
The second number is the first plus 236: 181 + 236 = 417 and 149 + 236 = 385. Likewise 262 + 236 = 498.

Q.5 · Answer (B) PEJKAQA
Consonants move 3 places forward and vowels stay: R → U, M → P, S → V, H → K in RAMESH → UAPEVK. So MEGHANA → PEJKAQA.

Q.8 · Answer (C) UAGL
The first three letters move 5 places forward and the last one 4 (wrapping round after Z): EKQW → JPVA and BHNT → GMSX. So PVBH → UAGL.

Q.9 · Answer (D) 134
The differences are 3, 6, 9 and 12, rising by 3, so the next is 15: 119 + 15 = 134.

Q.10 · Answer (B) 5
After interchanging the signs: 13 – 4 ÷ 2 × 5 + 2 = 13 – 10 + 2 = 5.

Q.11 · Answer (B) (16, 21, 672)
In each set the third number is twice the product of the first two: 25 × 8 × 2 = 400 and 12 × 14 × 2 = 336. Likewise 16 × 21 × 2 = 672.

Q.12 · Answer (A) Only conclusion I follows
All hens are chickens and all chickens are eggs, so all hens are eggs (I). Nothing says every egg is a chicken, so II does not follow.

Q.13 · Answer (A) 13
After interchanging the signs: 22 ÷ 11 – 13 + 12 × 2 = 2 – 13 + 24 = 13.

Q.14 · Answer (D) RMXY
The letters move +5, –3, +5, –3, wrapping round the alphabet: ARVL → FOAI and DOXY → ILCV. So MPSB → RMXY.

Q.15 · Answer (B) Murli
Facing the centre, right is anticlockwise. Virat is on Jadeja's right, Sarfaraz between Virat and Murli, and Ashwin beside Jadeja, so clockwise they run Jadeja, Ashwin, Karthik, Murli, Sarfaraz, Virat. Murli sits between Karthik and Sarfaraz.

Q.16 · Answer (D) er
'barks' is the only word in both sentences and 'er' the only code in both, so barks = er.

Q.18 · Answer (C) CTG
First letters move back 4 (S, O, K, G, so C), second letters forward 4 (D, H, L, P, so T) and third letters back 4 (W, S, O, K, so G): CTG.

Q.19 · Answer (C) 7
M is the only letter in both words and 7 the only digit in both codes, so M = 7.

Q.20 · Answer (A) Only conclusion [1] follows
Wolves are kept apart only from possums, so all wolves could be bisons: [1] follows as a possibility. Some bisons are possums and no possum is a wolf, so not every bison is a wolf: [2] does not follow.

General Awareness

Q.1 · Answer (B) My Country My Life

'My Country My Life' (2008) is L. K. Advani's autobiography. 'Wings of Fire' is A. P. J. Abdul Kalam's, and 'Exam Warriors' is by Narendra Modi.

Q.2 · Answer (D) 228.9 million

The 2022 global Multidimensional Poverty Index (UNDP and OPHI) put India's poor at 228.9 million in 2020, the largest number of any country.

Q.3 · Answer (B) 34

With two new judges sworn in on 13 February 2023, the Supreme Court reached its full sanctioned strength of 34, including the Chief Justice.

Q.4 · Answer (C) Debendranath Tagore

Debendranath Tagore, who revived the Brahmo Samaj after Raja Ram Mohan Roy's death, wrote the Brahmo Covenant.

Q.5 · Answer (D) Agriculture

Irrigation takes most of India's groundwater: close to 90% of what is drawn goes to agriculture.

Q.6 · Answer (C) Lord William Bentinck

Lord William Bentinck, Governor-General in the 1830s, wrote these words about the ruin of India's hand-weavers, undercut by machine-made British cloth.

Q.7 · Answer (A) Sheetal Devi

Archer Sheetal Devi won two golds at the Hangzhou Asian Para Games in October 2023, in the women's individual compound and the mixed team compound.

Q.8 · Answer (D) Sugarcane

Sugarcane takes about a year, 10 to 18 months, to mature; ragi, groundnut and soyabean are harvested within a few months of sowing.

Q.9 · Answer (B) Dhrupad

Ustad Wasifuddin Dagar is a dhrupad singer of the Dagar family, the best-known lineage of dhrupad.

Q.10 · Answer (D) Odisha

The International Sand Art Festival is held on Chandrabhaga beach near Konark, Odisha.

Q.11 · Answer (C) Fungus

Fungi and bacteria are decomposers: they break down dead matter and return its nutrients to the soil.

Q.12 · Answer (A) Bihar

Sama-Chakeva is a folk festival and dance of the Mithila region of Bihar.

Q.13 · Answer (C) Atomic radius decreases

Across a period the nuclear charge rises while electrons enter the same shell, so they are pulled closer and the atomic radius decreases.

Q.14 · Answer (D) Tamil Nadu

Tamil Nadu received the Sportstar ACES award for the best state for the promotion of sport in February 2024; it had hosted the men's hockey Asian Champions Trophy and the Khelo India Youth Games.

Q.15 · Answer (A) JM Keynes

John Maynard Keynes argued that government spending and taxation should be used to stabilise output and employment.

Q.16 · Answer (A) Inamgaon

Inamgaon, in Pune district of Maharashtra, lies on the Ghod, a tributary of the Bhima.

Q.17 · Answer (C) Garbha Griha
The garbhagriha, the 'womb chamber', is the sanctum that houses the main deity; the mandapa is the hall before it and the shikhara the tower above it.

Q.18 · Answer (C) Ag_2S
Silver reacts with sulphur compounds in the air to form black silver sulphide, Ag_2S: the tarnish on silver jewellery.

Q.19 · Answer (A) Vision 2035
NITI Aayog's white paper 'Vision 2035: Public Health Surveillance in India' sets out how India's disease surveillance should develop.

Q.20 · Answer (C) Albert Venn Dicey
The rule of law was set out by the British jurist A. V. Dicey: no one is above the law, and all are equal before it.

Q.21 · Answer (C) Goa
Dudhsagar Falls, on the Mandovi at Goa's border with Karnataka, is one of India's tallest waterfalls.

Q.22 · Answer (D) February 1999
Atal Bihari Vajpayee and Nawaz Sharif signed the Lahore Declaration on 21 February 1999, after Vajpayee's bus journey to Lahore.

Q.23 · Answer (C) Tanka
Iltutmish introduced the silver tanka and the copper jital, the standard coins of the Delhi Sultanate.

Q.24 · Answer (A) 2016
Gulabo Sapera, who took the Kalbeliya dance to the world, received the Padma Shri in 2016.

Q.25 · Answer (B) Concurrent List
The Seventh Schedule divides legislative powers among the Union List, the State List and the Concurrent List, on which both Parliament and the states may legislate.

English Language and Comprehension

Q.1 · Answer (B) dearest
'The' before the adjective calls for the superlative: 'Commodity prices are the dearest'.

Q.2 · Answer (D) Teetotaller
A teetotaller never drinks alcohol. A termagant is a quarrelsome woman and a Samaritan a helpful stranger.

Q.3 · Answer (D) The students were diligently
An action that went on for weeks up to a point in the past takes the past perfect continuous: 'The students had been diligently studying'.

Q.4 · Answer (C) dangerous
The correct spelling is 'dangerous'.

Q.5 · Answer (B) is the colder
Comparing one with all the states needs the superlative: 'is the coldest of all the states'.

Q.6 · Answer (D) in America brought a
A country takes 'in', not 'at': 'His aunt who lives in America'.

Q.7 · Answer (A) the best
SSC's key gives 'the best', treating the superlative as needing 'the': 'Which flower do you like the best?' 'Like best' without 'the' is also standard English, so the sentence as printed is arguably correct already.

Q.8 · Answer (A) restored
'Ravaged' means badly damaged; its opposite here is 'restored', brought back to a good condition.

Q.9 · Answer (C) has being
'Has being' is not a verb form; the sentence needs 'is' (or 'has been'): 'this new collection of paintings is a masterpiece'.

Q.10 · Answer (D) Incredible
'Incredible' means impossible or hard to believe; 'exquisite' means extremely beautiful.

Q.11 · Answer (D) Retrospect
To look back on one's past is to 'retrospect'; to 'introspect' is to examine one's own thoughts and feelings, and a 'prospect' looks ahead.

Q.12 · Answer (B) exhausted
Someone 'blue in the face' is worn out by effort, as after a long run. 'Feeling blue', a different idiom, means sad.

Q.13 · Answer (C) Aresst
The correct spelling is 'arrest'.

Q.14 · Answer (A) No substitution required
An unreal past condition takes the past perfect in the 'if' clause and 'could have' in the main clause, so the sentence is correct as it stands.

Q.15 · Answer (B) accomplish
To 'execute' a task is to carry it out: to 'accomplish' it.

Q.16 · Answer (C) Dissuade and Encourage
To 'dissuade' is to persuade someone not to do something, the opposite of 'encourage'. The other pairs are near-synonyms.

Q.17 · Answer (D) Bold
'Timid' means lacking courage; its opposite is 'bold'.

Q.18 · Answer (A) Revealed a secret or disclosed information
To 'spill the beans' is to reveal a secret.

Q.19 · Answer (C) enthusiasm
'Excitement' here is eager interest: 'enthusiasm'.

Q.20 · Answer (B) expect
The correct spelling is 'expect'.

Q.21 · Answer (C) featuring
A brigantine is a vessel 'featuring', that is having, fore-and-aft rigging on the mainmast.

Q.22 · Answer (A) ravaged
Pirates 'ravaged', or plundered, the Mediterranean.

Q.23 · Answer (D) evolved
The brigantine 'evolved into' a true sailing vessel: it developed gradually.

Q.25 · Answer (A) instance
'For instance' introduces an example; 'for case' is not an English phrase.

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